

Heartbeat-Evoked Potentials Reveal a Sleep-Stage and Age Gradient in Cortical Interoception

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Introduction

The heartbeat-evoked potential (HEP) is an R-peak-locked EEG deflection reflecting cortical processing of cardiac afferent signals linked to interoception and self-referential processing,^{5,9,10} contaminated by the cardiac field artifact; the –50 to +50 ms window around the R-peak is excluded from every test, per current HEP methodological guidance.⁷ Prior work found HEP magnitude follows a vigilance-state gradient across sleep (REM > light sleep [LS] > deep sleep [DS])² and increases with age,^{3,4} consistent with broader evidence of dynamic central-autonomic coupling across sleep.⁸ Whether these effects share a generator, and their topography, has not been tested.

Methods

Data were drawn from the Human Sleep Project, a multi-centre polysomnography corpus of 90,166 patients; 2,443 had the full 19-electrode montage, a heterogeneous referral cohort. This multi-hospital design parallels prior multicentre epilepsy imaging work.¹ Sleep-stage contrasts were tested with paired cluster-mass permutation tests, following the nonparametric cluster-based framework for EEG/MEG data,⁶ using 200 permutations, $\alpha = 0.01$; an Older-vs-Younger contrast (median split) was run within REM, each tested electrode-by-electrode.

Results

REM – DS showed a robust post-QRS difference (N = 2136, $p = 0.0050$; 15/19 electrodes), as did Older – Younger REM (N = 1144/1299, $p = 0.0050$; 9/19 electrodes). REM – LS was smaller, missing threshold (N = 2202, $p = 0.0149$; 1/19 electrodes). All three peaked outside the excluded CFA window, indicating genuine response.

Conclusion

The topography reveals a two-tier gradient: REM and LS barely differ, while both diverge sharply from DS, tracking arousability more than sleep depth — REM and LS preserve cortical reactivity, DS reflects maximal deafferentation and vagal dominance. The REM age effect matches the DS contrast in magnitude, suggesting ageing shifts interoceptive gain along the sleep-depth axis. Comparisons not age-matched, or pooling REM with LS with DS, risk mistaking a physiological shift for disease.

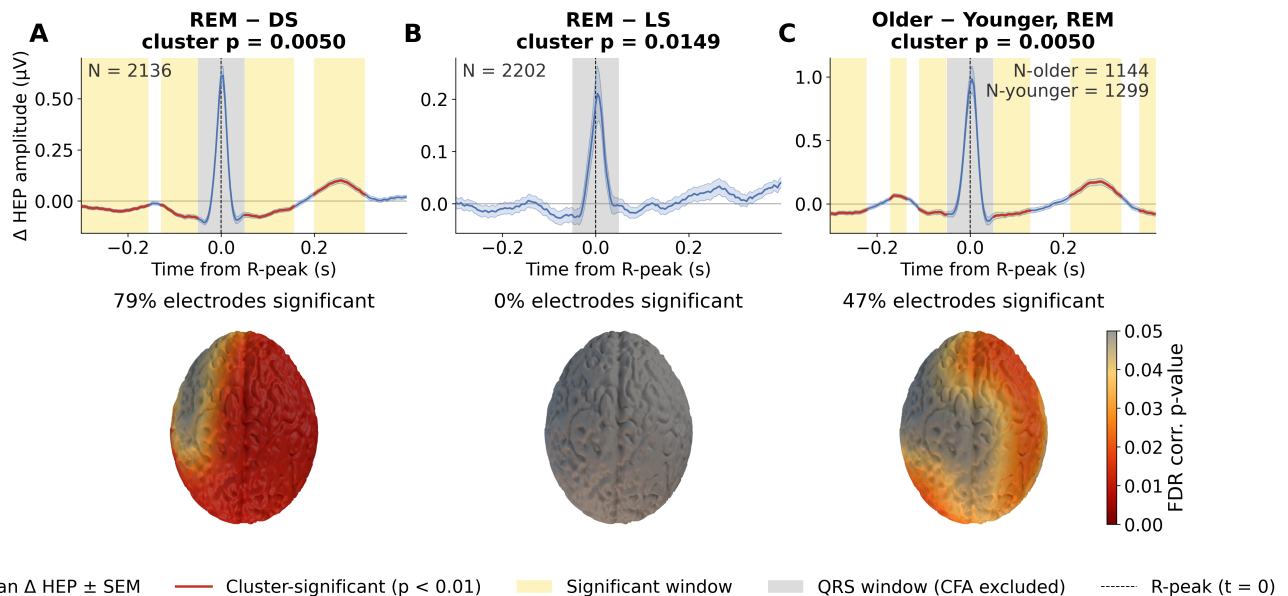


Figure. Each panel: mean Δ HEP waveform (left) and per-electrode significance topomap (right). Red = cluster-significant ($p < 0.01$); grey = CFA-excluded window. **A:** REM – DS. **B:** REM – LS (within-subject, paired, Susp. Epilepsy cohort). **C:** Older – Younger, REM (age median-split).

Acknowledgements

The Human Sleep Project has received support from the Glenn Foundation and the American Federation of Aging Research (AFAR) through the 2018 Glenn / AFAR Award for Medical Research Breakthroughs in Gerontology (BIG) (2018), the American Academy of Sleep Medicine (AASM) through a 2019 Strategic Research Award, the National Institutes of Health (NIH) (R01NS102190, R01NS102574, R01NS107291, RF1AG064312, RF1NS120947, R01AG073410, R01HL161253, R01NS126282, R01AG073598), the National Science Foundation (NSF 2014431), and through the Henry and Allison McCance Center for Brain Health.

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